

Tunis, Tunisia

+216 22 296 652

maramnuiguess@gmail.com

linkedin.com/in/maram-nouiguez-ab929a294

Key Skills: Lean Management • VS Matrix / VSM / VSD • 5S • Gemba Walk • Root Cause Analysis • Line Balancing • LP/MILP Optimization • Power BI / Excel VBA

Professional Profile

Final-year (5th year) Industrial Computing and Automation student at INSAT, trained on the plant floor in Lean tools, flow analysis, and industrial planning.

At Autoliv, I built (from scratch, in Excel first, then as a web app) the tool used to turn new products into equipment plans: a process that used to take a cross-functional team **more than 4 working days** now takes **under 20 minutes for one person**.

Looking for a final-year internship in Continuous Improvement, Lean Management, or Industrial Planning & Supply Chain, starting February 2027.

Education

Engineering Diploma, Industrial Computing and Automation – Intelligent Systems track

2022 – 2027

INSAT – National Institute of Applied Sciences and Technology, Tunis (Machine Learning, Big Data, Deep Learning, Data Science, Operations Research, UML)

Professional Experience

Intern, APS Department (Autoliv Production System – Continuous Improvement & Digitalization)

Jun – Aug 2025

Autoliv, El Fahs, Tunisia – supervised by Slim Hsan, APS Manager

- VS Matrix ownership:** learned and applied the VS Matrix method – grouping new products into flows/value streams, sizing equipment per flow, and deciding on new equipment or CAPEX to support one-piece-flow and line balancing.
- Process automation:** first mapped the entire method manually in Excel, then built a **Python/Streamlit** app (with an AI-assisted compatibility matrix) to run it, plus a separate **HTML/JS drag-and-drop** factory-layout simulator – cutting a task that needed **4+ working days from a cross-functional team** (process, launch, logistics, quality) down to **under 20 minutes for a single person**.
- Capacity planning:** sized equipment per flow based on a 3-shift, 20-day/month capacity model with a < 85% utilization target.
- Rebuilt the plant's **5S Lean dashboard in Python (openpyxl)**, with automated KPI cards, charts and formula-driven aggregation; also studied the factory layout in AutoCAD and picked up L2L and PLM.
- VSM workshop, high-runner product:** joined the team investigating recurring issues found in an internal audit; ran a Gemba walk, then built the VSM, VSD and VSD². My part: consolidating the team's findings in draw.io, building the ABC matrix, and preparing the final presentation.
- Invited to replicate the VS Matrix exercise at Autoliv's Nadhour plant.

Intern, Industrial Engineering

2024

Lacroix Electronics, Tunisia

- Helped diagnose a malfunctioning robot arm using Lean root-cause tools; traced the issue to a sensor sending faulty signals to the PLC – a first hands-on look at where Lean thinking and automation meet.

Technical Projects

Operations Research – Company Logistics & Distribution Study – modeled a company's production, transport and product-mix planning using LP/MILP, with sensitivity analysis to support efficient distribution decisions.

AnyLogic Manufacturing Supply Chain Simulation – built a custom simulation of a manufacturing supply chain; tracked KPIs and identified bottle-necks, with results linked back to Excel.

Academic Project Management – ran a project end to end: charter, stakeholder register/RACI, WBS, PERT/CPM/Gantt, 5-year Life Cycle Costing budget, risk register, and a KPI/ESG dashboard.

Additional projects: SAP production planning, Factory IO/TIA Portal automation, embedded vision quality control, industrial digital twin, autonomous inspection robot, VoltMind AI assistant, Spark (educational STEM app), early line-follower robot.

Technical Skills

Lean & Continuous Improvement:

VS Matrix, VSM/VSD/VSD², 5S, Six Sigma, Gemba walk, ABC matrix, 5 Why, 8D, Ishikawa, SMED, AMDEC, line balancing, one-piece-flow

Planning & Data:

LP/MILP modeling, Power BI, Excel (VBA, pivot tables), Python (Pandas, Streamlit, openpyxl), SQL

Digitalization Tools:

L2L, PLM, AutoCAD, draw.io

Languages:

Arabic (native), French (C1), English (C1)

Leadership & Distinctions

Chairwoman, IEEE Women in Engineering – INSAT SAG (2024–2025) – led a **100+ member** organization: budget, volunteer team, and end-to-end technical events.

Team Lead, VIP & Protocol Department – National Robotics Week 8.0 (2026) – ran the VIP & Protocol team and developed panel content (automotive/SDV, cyber-physical security) for a national robotics event.

Robotics Instructor, Play Mind Club (2025 – Jun 2026) – ran STEM workshops (Scratch, Arduino, robotics) for children aged 8–12.